Conjecture 00000002894: a vertexless triangle in PG(2, 9)

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23 October 2026

1 The classification fails without a smallness restriction

The English statement claims that nontrivial minimal blocking sets in planes of order q^2 are of Rédei type, and that non-Rédei examples exist only when q is square. Neither clause imposes a smallness restriction. We give an inclusion-minimal counterexample in the Desarguesian plane of order $Q = 9 = 3^2$. Its parameter $q = 3$ is not square. Here "minimal" means inclusion-minimal, as in the English statement and standard terminology; it does not mean minimum cardinality.

A blocking set meets every line. It is nontrivial if it contains no whole line, and minimal if no proper subset is blocking. A Rédei blocking set has a line ℓ with $|B \setminus \ell| = Q$. These definitions appear in Gács–Lovász–Szőnyi, *Directions in* $\mathrm{AG}(2, p^2)$, Innovations in Incidence Geometry 6–7 (2007–08), pp.189–201, printed p.191. In particular, the necessary property of Rédei type used here does not require an additional size bound.

2 The actual point set and its tangents

Work over $K = \mathbf{F}_9$, in homogeneous coordinates. Let

$$B = \{[0 : 1 : t], [t : 0 : 1], [1 : t : 0] : t \in K^\times\}.$$

These are precisely the three coordinate sides with their three vertices removed. The three families are disjoint, and each has 8 distinct points, so $|B| = 24$.

Every nonzero linear form $a_0x + a_1y + a_2z$ vanishes at some point of B. If two coefficients are nonzero, choose the corresponding coordinates to cancel and set the third coordinate to zero. If just one coefficient is nonzero, choose its coordinate to be zero and the other two to be one. Thus B is blocking.

The tangent to $[0 : 1 : t]$ has equation $-ty + z = 0$; cyclically the other tangents are $x - tz = 0$ and $-tx + y = 0$. Each meets B in exactly the indicated point. Removing any point therefore leaves a line unblocked. More generally, any blocking subset of B must contain every point of B, which proves inclusion-minimality.

3 Nontriviality and failure of Rédei type

Every line has a point outside B. If a coefficient a_i of its nonzero linear form is zero, the corresponding coordinate vertex lies on the line and is outside B. Otherwise the point

$$[a_0^{-1} : a_1^{-1} : a_2^{-1}]$$

lies on the line because $1 + 1 + 1 = 0$ in characteristic three. All its coordinates are nonzero, so it is outside B. Hence B contains no line and is nontrivial.

A projective line over K has $Q + 1 = 10$ points. Consequently every line satisfies

$$|B \setminus \ell| = 24 - |B \cap \ell| \ \geq 24 - 10 = 14 > 9.$$

There is no Rédei line. Thus this same actual set refutes both the universal Rédei assertion and the claimed square-q restriction. The further characterization of Rédei sets cannot repair either false necessary clause and is not needed for the disproof.

4 Formal correspondence

The Lean point type is Mathlib's genuine `Projectivization K (Fin 3 → K)`, not a list of assumed incidence objects. Lines are the projectivizations of kernels of all nonzero homogeneous linear forms. The kernel dimension is proved to be two by surjectivity of the form and rank–nullity. A restriction map into the projectivization of that kernel is proved injective, giving the line cardinality upper bound from Mathlib's finite projective-space cardinality theorem. Only this upper bound is needed in the formal contradiction.

The point set is the image of the explicit three families, and the family map is proved injective using projective scalar equivalence. Blocking, the exact tangent intersections, minimality and nontriviality are proved from coordinates. The formal predicate `IsRedei` is exactly the displayed necessary line-complement criterion. The final `counterexample` specializes to `GaloisField 3 2`, proves its cardinality is $3^2 = 9$, proves 3 is not square, and bundles the 24-point count and all four blocking/minimal/nontrivial/non-Rédei properties. No stochastic or numerical incidence test replaces the mathematical objects.

5 Verification

The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

Source-bound receipts accompany a new Main build, transitive axiom audits and official `leanchecker --fresh Main` replay. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed the formal chain. This is AI review, not human certification. Organizer acceptance is pending.